Secure Healthcare FL Layered Architecture

Governance, local data, collective keys and privacy-safe training

Secure Multi-Institutional Healthcare FL System

L0 Governance, Consent and Compliance

Consortium CA
Hospital identity and roles

Consent Registry
purpose, scope, expiry, revocation

C

Governance Policy
retention, access and incident rules

Policy

L1 Local Healthcare Data Layer

Hospitals A, B, C ... N

Consent and purpose check

Local preprocessing
feature extraction and normalization

Local healthcare dataset
raw data never leaves hospital

L2 Distributed Cryptographic Setup

Key rotation and revocation

Distributed key generation

Collective CKKS public key

Threshold secret-key shares
t-of-n hospitals

Key shares protected by
HSM or secure enclave

B

Public key

L3 Privacy-Safe Local Training

D

Local model training

Per-client gradient
clipping

Gaussian DP noise

Privacy accountant
epsilon and delta

DP-safe adaptive
compression

Padded sparse encoding
hide data-dependent
payload size

CKKS encryption

Digital signature + round ID
nonce and timestamp

A

Cross-panel interfaces: A signed update -> TLS B key shares -> threshold decryption C consent registry -> inference check D next round -> local training

Secure Healthcare FL Layered Architecture

Secure transport, resilient aggregation, verified distribution and clinical inference

L4 Secure Communication and Trust

A

Mutual TLS secure channel

Verify certificate, signature, nonce and round freshness

Smart-contract admission identity and reputation policy

Clipping-bound proof or trusted execution attestation

Encrypted updates stored off-chain

Permissioned ledger stores only hash, signature, timestamp and status

Ciphertexts + ledger status

L5 Secure and Poisoning-Resilient Aggregation

B

Minimum-client quorum dropout-tolerant collection

Reputation-aware weighted homomorphic FedAvg

t-of-n threshold decryption aggregate only

Candidate model validation quality, drift and anomaly checks

Validation passed?

Yes

Accept global model

No

Reject round, rollback and quarantine suspects

Update reputation and audit evidence

Accepted model

L6 Verified Distribution

Record global-model hash

Signed secure broadcast

Hospitals verify version and hash

Begin next FL round

D

Accepted model

L7 Privacy-Preserving Clinical Inference

C

Authenticated encrypted request

Consent and authorization check

Encrypted or confidential inference

Threshold or enclave decryption prediction result only

Output minimization, rate limiting and audited clinical decision support

Cross-panel interfaces: A signed update -> TLS B key shares -> threshold decryption C consent registry -> inference check D next round -> local training